

Joshua Gottus

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U.S. Citizen | Security Clearance Eligible

EDUCATION

Florida Polytechnic University

B.S. Computer Science, Cybersecurity Concentration

Expected May 2027

Lakeland, FL

South Florida State College

Associate of Arts

May 2024

Avon Park, FL

EXPERIENCE

VES LLC

Embedded Software Engineering Intern

May 2026 – Aug 2026

Tampa, FL

- Led design of a distributed multi-vehicle autonomy system (1 of 4 interns, 10-person team), bridging ArduPilot to ROS 2 via MAVROS, with inter-vehicle traffic on Zenoh over Silvus MANET radios.
- Developed a C++ ROS 2 node that built costmaps from SLAM Toolbox maps and FAST-LIO point clouds for Nav2 and EGO-Swarm, with vehicle voting on goals for GPS-denied operation.
- Ported Python prototypes to C++ ROS 2 nodes on Raspberry Pi and Jetson Orin Nano companion computers.
- Designed the UGV electrical system in KiCad (schematics, power distribution, wiring harnesses, component selection) and delivered the design to the electrical engineer for review and build.
- Cut vehicle container image size 82% (5 GB to 0.9 GB), reducing deployment time to onboard computers.
- Validated UGV and UAV autonomy in ArduPilot SITL before a UGV field test.
- Wrote a design document per story, revised it through tech lead and peer review, and reviewed team PRs across 100 stories and subtasks.

Florida Polytechnic University

IT Help Desk Student Technician

Jul 2025 – May 2026

Lakeland, FL

- Administered Linux and Windows servers and endpoints across networking, identity management, and system maintenance.
- Automated endpoint provisioning and software deployment with Bash and PowerShell through SCCM and Intune.
- Coordinated with IT staff to resolve support tickets and complete software rollouts across campus endpoints, scoring 5 of 5 stars on 30+ rated tickets.

Highlands County Clerk of Courts

IT Technician

Jun 2023 – Aug 2023

Sebring, FL

- Automated system administration tasks with Python, PowerShell, and shell scripts.
- Resolved help desk tickets for county court staff and systems.

PROJECTS

Flight Control Data Bus | C, FreeRTOS, ROS 2, Micro-ROS

Sep 2026 – Present

- Designed and implemented multi-node flight control firmware in C on FreeRTOS
- Implemented two level watchdog supervision for software task health monitor gating hardware watchdog
- Enforced safety critical practices such as static memory allocation only, fixed point math and zero warning builds

SoutheastCon 2026 Ground Robot | C, Linux, Raspberry Pi, I2C

Jun 2025 – Apr 2026

Software Tech Lead and Project Manager

github.com/Florida-Poly-IEEE-RAS-Society/SECON26

- Led a 9-person software team to 16th of 84 at IEEE SoutheastCon 2026, writing and assigning GitHub issues and reviewing team PRs before merge.
- Developed modular C software on Raspberry Pi with a hardware abstraction layer over an I2C motor driver and IMU, running a 100 Hz control loop.
- Used an oscilloscope to trace a failed Raspberry Pi 5 to a 5 V signal driving a 3.3 V GPIO pin, then replaced the board.

MicroTransit GPS Tracker | C++, Arduino, SIM7000A, MQTT, FastAPI, Next.js

Mar 2026 – May 2026

- Co-developed C++ firmware with an electrical engineering teammate for an Arduino Uno (2 KB SRAM) and SIM7000A modem, with a 46-entry EEPROM offline buffer, layered auto-reconnect, and an 8 s hardware watchdog.
- Built a FastAPI backend (MQTT, Kalman-filtered GPS, SQLite, WebSocket) and Next.js live map on Docker Compose, and taught the teammate each layer of the stack.

TECHNICAL SKILLS

Languages: C, C++, Python, TypeScript

Embedded: FreeRTOS, micro-ROS, STM32, ESP32, Arduino, Raspberry Pi, Jetson Orin Nano, UART, I2C, PWM, GPIO

Robotics & Autonomy: ROS 2, Nav2, SLAM Toolbox, FAST-LIO, EGO-Swarm, ArduPilot SITL, MAVLink, MAVROS

Communications: Zenoh, Silvus MANET Radios, MQTT, WebSocket, LTE-M (SIM7000A)

Tools: Linux, Git, Docker, CMake, GitLab CI/CD, GitHub Actions, Bash, PowerShell, KiCad, Oscilloscope